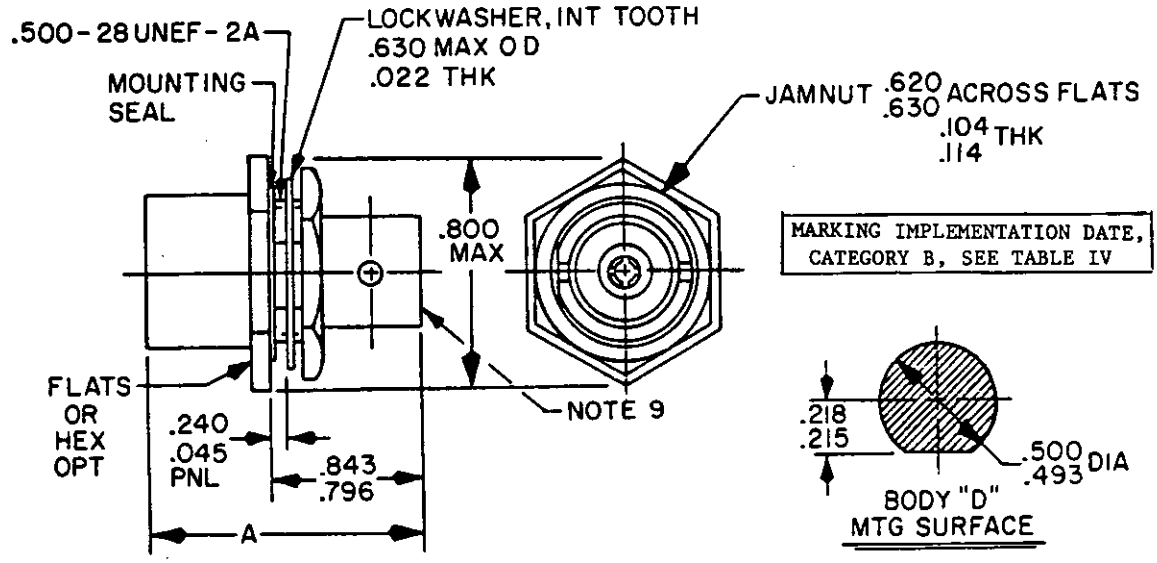
MILITARY SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES BNC (CABLED), SOCKET CONTACT, JAM NUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



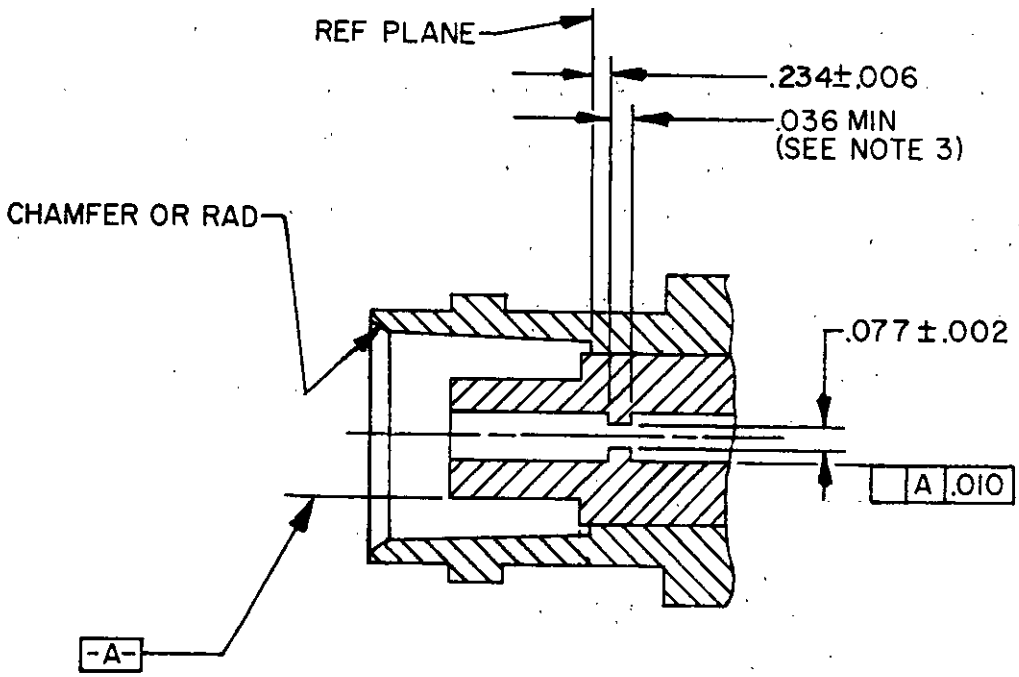
Inches	mm	Inches	mm
.022	0.56	.493	12.52
.045	1.14	.500	12.70
.104	2.64	.505	12.83
.114	2.90	.510	12.95
.215	5.46	.620	15.75
.218	5.54	.630	16.00
.221	5.61	.796	20.22
.240	6.10	.800	20.32
		.843	21.41

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For dimension A, see table I and III.
4. Dimension .800 (20.32 mm) is the largest overall diameter of the connector.
5. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28, appendix 10.
6. Orientation of the body hex flats, mounting flats and bayonet studs shall be ±3° of that orientation shown.
7. All undimensioned pictorial representations are for reference purposes only.
8. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
9. Series BNC, socket contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A
FSC 5935

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Inches	mm
.002	.05
.006	.15
.010	.25
.036	.91
.077	1.96
.234	5.94

CATEGORY D

NOTES:
1. Metric equivalents are given for general information only.
2. Concave depression .100 (2.54 mm) x .005 (0.13 mm) deep between studs permissible.
3. Chamfer is optional.
4. Dimensions are in inches.

FIGURE 2. Category D captivation detail.

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TABLE I. Dash numbers, cross reference, and dimensions.

Dash number 1/	Applicable cable M17/#	Dimensions	Inches (millimeters) maximum
Category A - Field serviceable (no special tools required)			
0101 4/	028-RG58 084-RG223* 128-RG400 060-RG142@	A	1.250 (31.75)
0102 4/	029-RG059*Δ 030-RG062Δ 90-RG71Δ 97-RG210Δ		
0110 4/	054-RG122*		
0111 4/	111-RG303@*		
0118 4/	110-RG302*@Δ		
0220 4/	113-RG316@* 119-RG174		
Category C - Field replaceable (MIL-C-22520 crimp tool) 3/ See footnote next to applicable cable for crimp die			
0013 4/	028-RG058*~ 111-RG303~	A	1.625 (41.28)
0014 4/	084-RG223*~ 060-RG142~		
0015 4/	29-RG59*Δ> 030-RG062Δ> 97-RG210Δ>		
0016 4/	054-RG122*&		
0017 4/	90-RG71*Δ>		
0020 4/	110-RG302*@Δ>		
0221 4/	113-RG316@Σ 119-RG174Σ		
Category D - field replaceable - Defined piece parts 3/5/2/			
0501 4/	054-RG122*	A	1.468 (37.31)
0502 4/	095-RG180*Δ		
0503 4/	060-RG142* 128-RG400		
0504 4/	028-RG058*		

See footnotes on following page.

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1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ Not to be used in Army equipment.

3/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

4/ These connectors have captivated center contacts.

5/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and ∆.

@ Cable to be used for the +200°C temperature cycling tests.

∆ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

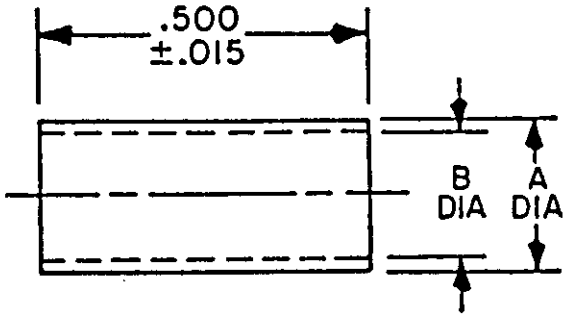
Σ M22520/5-35 Closure B or M22520/5-03 Closure A

& M22520/5-41 Closure B or M22520/5-05 Closure B
M22520/5-09 Closure A

~ M22520/5-19 Closure B or M22520/5-05 Closure A
M22520/5-11 Closure A
M22520/5-57 Closure A

> M22520/5-19 Closure A or M22520/5-07 Closure A
M22520/5-13 Closure A
M22520/5-59 Closure A

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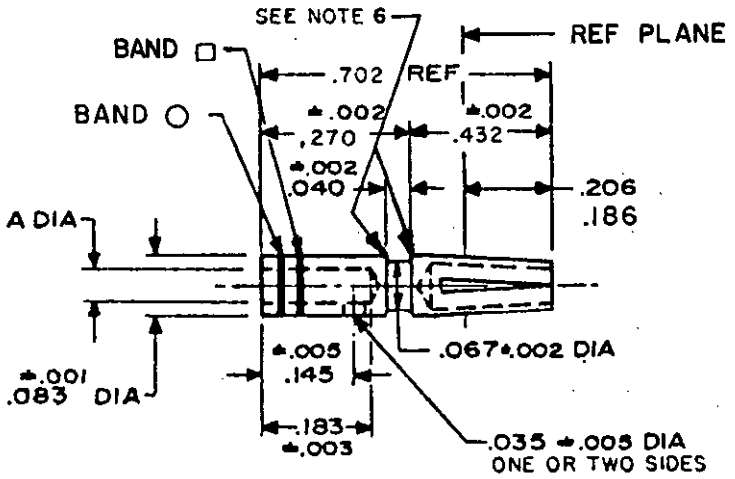


Dash No.	Ferrule No. 2/	A ±.001	B ±.001	Basic crimp tool 1/	Crimp die or positioner M22520/5-
0501 0502	19-50	.212	.175	M22520/5-01	05, 41 Closure B or 9 Closure A
0503	19-51	.250	.220		05, 11, 57 Closure A or 19 Closure B
0504	19-52	.245	.206		

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 3. Contact and ferrule dimensions for category D only.

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CENTER CONTACT

Inches	mm
.001	.03
.002	.05
.005	.13
.015	.38
.017	.43
.033	.84
.035	.89
.040	1.02
.043	1.09
.067	1.70
.083	2.11
.145	3.68
.175	4.45
.183	4.65
.186	4.72
.206	5.23
.212	5.38
.220	5.59
.245	6.22
.250	6.35
.270	6.86
.432	10.97
.500	12.70
.702	17.83

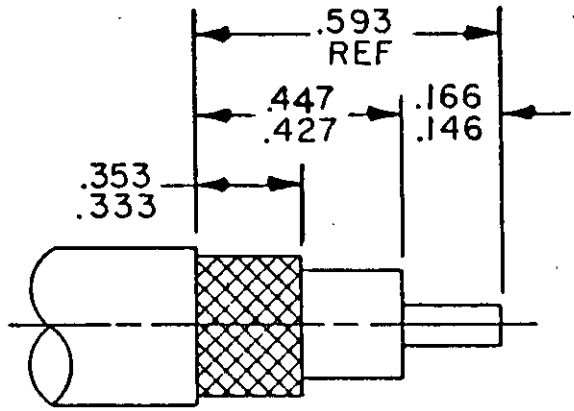
Dash No.	Contact No. 2/	A +.001 -.002	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501	19-12	.033	M22520/1-01	M22520/1-12	10 lbs	Orn.	Blue
0502	19-11	.017			6 lbs	Blue	
0503 0504	19-10	.043	M22520/1-01	M22520/1-12	20 lbs	Red	

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. .003 (0.08 mm) maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only.

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Inches	mm
.146	3.71
.166	4.22
.333	8.46
.353	8.97
.427	10.85
.447	11.35
.593	15.06

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 4,000 MHz.

Voltage rating:

500 volts rms, maximum working voltage at sea level.

125 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: 3 pounds, maximum.

Torque: 2-1/2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque not applicable.

Mating characteristics:

Reference MIL-STD-348 and figure 2 for dimensions.

Center contact (socket):

Oversize test pin: .057 diameter, minimum (nonclosed entry contacts only).

Insertion depth: .125, minimum.

Number of insertions: 1.

Insertion force test: Steel test pin diameter .054, minimum.

Test pin finish: 16 microinches.

Insertion force: 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .052, minimum.

Withdrawal force: 2 ounces, minimum.

Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage: Connector shall be mounted in mounting hole shown on figure 1 with mating end capped. Test applicable to mounting seal only.

Air pressure - 30 lbf/in^2.

Duration - 30 seconds, minimum.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

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Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 4,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30, maximum.

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.015+.005 F (F in GHz).

Item 16: VSWR shall be less than 1.015+.005 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.045+.019 F (F in GHz).

Group B inspection: VSWR shall be less than 1.1+.01 F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.15.

Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact	1.5	2.0
Outer contact	.2	Not applicable
Braid to body	.1	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 1,500 volts rms, minimum at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition G. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 375 volts rms, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,000 volts rms at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force:

Noncrimp assemblies: 40 pounds, minimum.

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Crimp assemblies:

10 pounds, minimum for cables .155 - .189 OD.

20 pounds, minimum for cables .190 - .229 OD.

30 pounds, minimum for cables .230 - .249 OD.

40 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -55 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: .2 dB maximum tested at 3 GHz.

Part number: M39012/19 (dash number from table I or "B" number from table III).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors M39012/	Qualifies the following connectors M39012/
I	17-0101 17-0103 17-0111 17-0118 18-0102 18-0103 18-0111 18-0118 19-0101 19-0110 19-0111 19-0118 19-0220	17-0101 17-0102 17-0103 17-0111 17-0118 18-0101 18-0102 18-0103 18-0111 18-0118 19-0101 19-0102 19-0110 19-0111 19-0118 19-0220
II	17-0102 18-0101 19-0102	17-0102 18-0101 19-0102
III	17B0004 17B0005 17B0006 17B0007 17B0009 17B0019 18B0004 18B0005 18B0006 18B0007 18B0009 18B0019 19B0003 19B0004 19B0005 19B0006 19B0008 19B0019	17B0004 17B0005 17B0006 17B0007 17B0008 17B0009 17B0012 17B0019 18B0004 18B0005 18B0006 18B0007 18B0008 18B0009 18B0012 18B0019 19B0003 19B0004 19B0005 19B0006 19B0007

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TABLE II. Group Qualification - Continued.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/	M39012/
III		19B0008 19B0012 19B0019
IV	17B0008 17B0012 18B0008 18B0012 19B0007 19B0012	17B0008 17B0012 18B0008 18B0012 19B0007 19B0012
V	17-0013 17-0014 17-0016 17-0020 18-0013 18-0014 18-0016 18-0020 19-0013 19-0014 19-0016 19-0020 19-0221	17-0013 17-0014 17-0015 17-0016 17-0017 17-0020 18-0013 18-0014 18-0015 18-0016 18-0017 18-0020 19-0013 19-0014 19-0015 19-0016 19-0017 19-0020 19-0221
VI	17-0015 17-0017 18-0015 18-0017 19-0015 19-0017	17-0015 17-0017 18-0015 18-0017 19-0015 19-0017
VII	19-0501 19-0503 19-0504	19-0501 19-0502 19-0503 19-0504
VIII	19-0502	19-0502

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only

Dash number 1/ 2/ 3/	Applicable cable M17/#	Dimensions	Inches (millimeters) maximum
M39012/19B0003	028-RG058*	A	1.625 (41.28)
M39012/19B0004	084-RG223*		
M39012/19B0005	111-RG303*		
M39012/19B0006	060-RG142@* 29-RG59*Δ 030-RG062Δ		
M39012/19B0007	97-RG210Δ		
M39012/19B0008	054-RG122*		
M39012/19B0012	90-RG71*Δ		
M39012/19B0019	110-RG302*@Δ		

1/ For cross reference of dash number to superseded part number or type designation, see table IV.

2/ For maintenance replacements for category B, see table V.

3/ Inactive for new design.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in notes @ and Δ.

@ Cable to be used for the +200°C temperature cycling tests.

Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross reference of part numbers.

Preferred part number M39012/19	Substitute for part number or type designation 1/2/
-0101	UG-909/U, M39012/19-0001
-0102	UG-910/U, M39012/19-0002
B0003 3/	M23329/3-15, M39012/19-0003
B0004 3/	M23329/3-16, M39012/19-0004
B0005 3/	M23329/3-17, M39012/19-0005
B0006 3/	M23329/3-18, M39012/19-0006
B0007 3/	M23329/3-19, M39012/19-0007
B0008 3/	M23329/3-20, M39012/19-0008
-0110	M39012/19-0010
-0111	M39012/19-0011
B0012 3/	M39012/19-0012
-0013	
-0014	
-0015	
-0016	
-0017	
-0118	M39012/19-0018
B0019 3/	M39012/19-0019
-0020	
-0501	
-0502	
-0503	
-0504	

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/19-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

3/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

TABLE V. Maintenance replacements for category B.

Category B number* Inactive for new design	Category C dash number	Category A dash number	Category D dash number
B0003	0013	0101	0504
B0004	0014	0101	---
B0005	0013	0101	---
B0006	0014	0101	0503
B0007	0015	0102	---
B0008	0016	0110	0501
B0012	0017	0102	---
B0019	0020	0118	---

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - CR, EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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